GUEST LECTURE • SPEECH RECOGNITION

Agentic AI in Speech Recognition

From recognizing speech to understanding, thinking, and taking action

Aisyah Izzah | Data Scientist, Fata Organa

Industry knowledge sharing session

Who am I?

A Data Scientist working at the intersection of speech, machine learning, and practical AI applications.

Speech & emotion AI

Exploring how machines can recognize information and patterns from human voice.

Machine learning

Developing models through data preparation, feature analysis, experiments, and validation.

Model evaluation

Checking performance, robustness, generalization, and where models make mistakes.

A key part of the job: turning experiments into reliable solutions

A little about Fata Organa

PT Fata Organa Solusi

Software and AI solutions for businesses.
A subsidiary of CAC Japan.

Software

Custom software, cloud and on-site systems.

AI applications

AI and automation for business tasks.

Our products in practice

After the concepts, we'll explore BelugaBox (EmoSDK), RTMT, and Speech-to-Speech AI.

Visit for more information: https://fata-organa.com/index.html

My work: developing and evaluating models for speech and emotion recognition.

What we'll explore

From recognizing speech to taking action—with lessons from industry.

02 Speech Recognition

How does a computer turn spoken words into text?

03–04 Agentic AI & Voice Agents

How can AI choose steps, use tools, and act on speech?

05 Products in Practice

BelugaBox / EmoSDK, RTMT, and Speech-to-Speech AI.

06–07 Industry Lessons & Outlook

How do we build reliable systems, and what comes next?

08 Hands-on Workshop

Build a speech-to-speech study agent in Python. Separate workshop guide.

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Can computers understand our voice?

Mentimeter Slide 2

Think about the voice technologies you already use.

Voice typing

Speak and see words appear

Virtual assistants

Ask a question or set a reminder

Video captions

Read what someone said

Voice conversations

Talk naturally with an AI

Question: what happens between sound and a useful response?

Speech Recognition

Speech Recognition enables computers to recognize the words we say and turn them into written text.

Spoken input

“Set a timer for five minutes.”

→

ASR

Text output

“Set a timer for five minutes.”

ASR answers: “What was said?”

How does Speech Recognition work?

It uses a trained model to recognize patterns in speech and predict words.

1 Audio input
Record the speech signal

2 Prepare audio
Format and segment the audio

3 Features
Convert sound into numerical features

4 ASR model
Predict a sequence of words

5 Text output
Display the recognized words

Simplified workflow: modern ASR systems may combine several stages in one model.

Training teaches the model to connect sound patterns with words.

Recognizing words is not the same as understanding meaning

Speech Recognition (ASR)

Input: “Can you open the window?”

Output: the words spoken, converted into text.

Language understanding

Possible meaning: a polite request to open the window — not a question about ability.

Context matters: “That’s just great…” can express genuine happiness or sarcasm.

How AI interaction is evolving

Command-based

Follow a predefined instruction

Set an alarm

Conversational

Answer questions and generate content

Explain a topic

Agentic

Plan and carry out multi-step tasks

Find a time and schedule a meeting

These are overlapping capabilities, not strict generations. A chatbot is not automatically an agent.

What is Agentic AI?

AI systems designed to work toward a goal by deciding what steps to take and using tools when needed.

Understand

What is the goal?

Plan

What steps are needed?

Use tools

Search, calculate, access systems

Check

Did the action work?

Example goal

“Help me plan a study week” → check availability → propose sessions → ask permission → add to calendar.

A simple agent workflow

If the result is not good enough, the agent may revise its plan and try again.

Teaching framework only: real implementations may combine or skip stages, and can still make mistakes.

What makes an AI agent work?

AI model

Interprets requests and decides what to do

Goal & instructions

Defines the task and boundaries

Tools

Search, database, calendar, APIs

Context / memory

Keeps relevant information

Orchestration

Coordinates the workflow

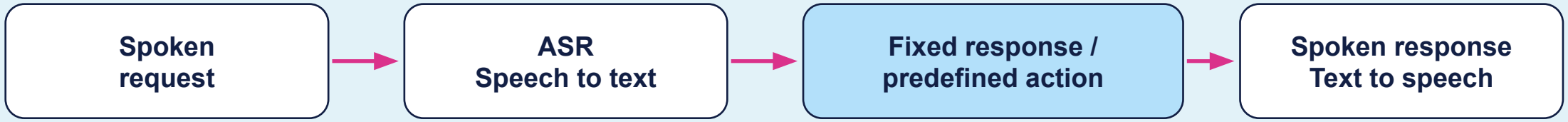
Guardrails

Permissions, validation, human approval

Important: tools let an agent interact with external systems. The model does not magically know live information.

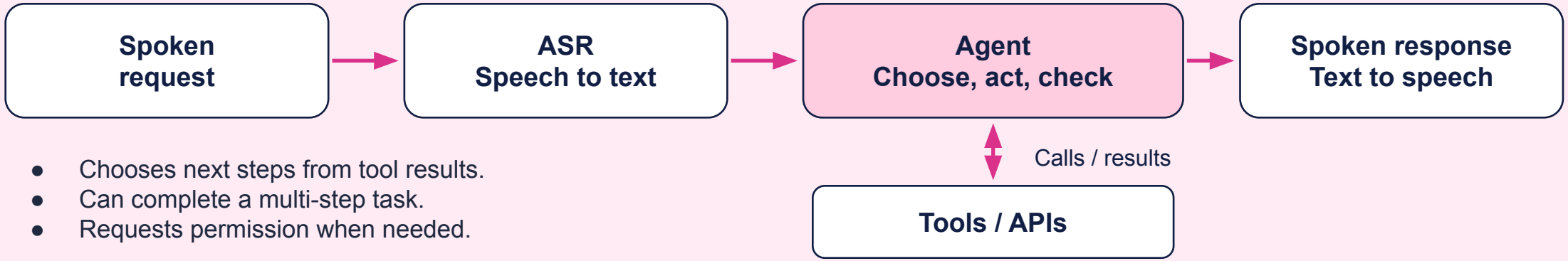
When Agentic AI meets Speech Recognition

Traditional voice assistant · predefined flow



- Follows configured steps to answer or carry out a predefined action.

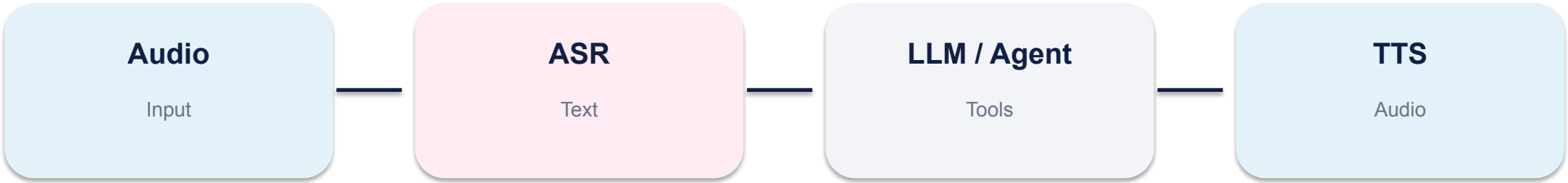
Voice-enabled agent · adaptive flow



- Chooses next steps from tool results.
- Can complete a multi-step task.
- Requests permission when needed.

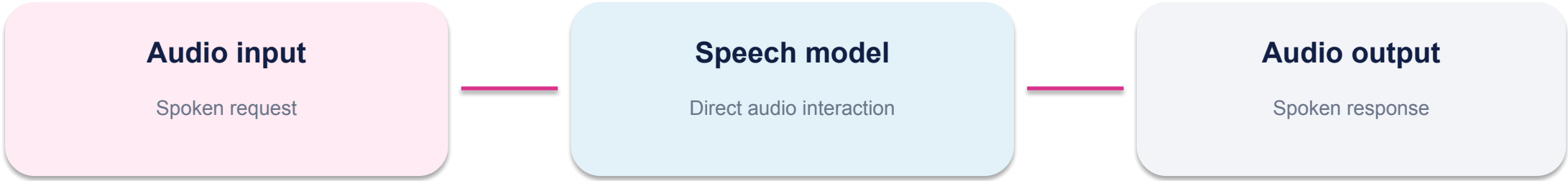
Two common voice-agent architectures

1. Chained architecture



Example: room-booking assistant
Speech → transcript → booking API → spoken confirmation.

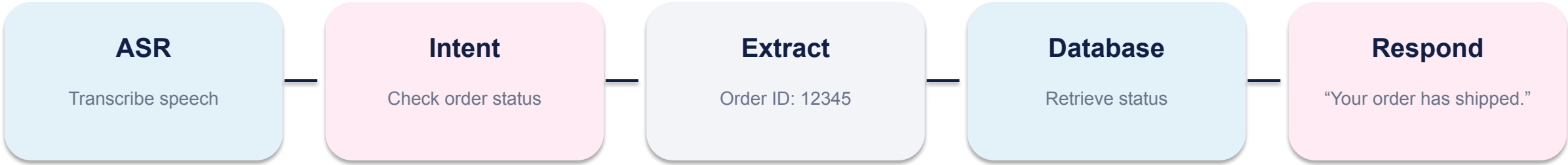
2. Speech-to-speech architecture



Example: live speaking partner
You speak in English; the model listens and replies in spoken English.

Example: a voice agent for customer service

Customer: “I want to check my order status. My order number is 12345.”



Where can things go wrong?

ASR may mishear 12345 as 12354. A wrong transcription can trigger a wrong database lookup.

Discussion: which steps belong to speech recognition, and which belong to the agent?

Real products: voice AI in everyday life

ChatGPT Voice

Natural spoken conversations; agentic capabilities depend on available tools and permissions.

Gemini Live

Real-time voice conversation and multimodal interaction.

Amazon Alexa

Voice commands for smart-home controls, routines, and information.

Customer-service voice agents

Voice interface + business data + permitted actions; can support multi-step workflows.

Key idea: a voice interface alone does not make a system agentic.

Mini game: ASR or Agent?

Mentimeter Slide 3 - 4

Vote for the capability responsible for each task.

1. Transcribe a recorded lecture into text.	ASR
2. Find a library book and reserve an available copy.	AGENT
3. Turn a classmate's voice message into text.	ASR
4. Check prerequisites and suggest electives you can take.	AGENT
5. Write down what you say during a presentation rehearsal.	ASR
6. Find open scholarships that match your student profile.	AGENT

ASR recognizes words; an agent uses information to choose actions.

How we apply these ideas at Fata Organa

Speech → Insights → Action

For our clients

Emotion insights, meeting support, and spoken access to company knowledge.

For our development team

Agentic AI supports experiments and data-quality checks before training.

Let's explore BelugaBox, RTMT, and Speech-to-Speech AI.

BelugaBox & EmoSDK: Emotion Insights

Recognized emotion becomes useful information for clients.

Emotion recognition

Acoustic features → ML model → emotion scores.

Client insights

Combine emotion scores with context to support client analysis.

Agentic AI integration

Use recognition outputs to guide further analysis and follow-up.

Illustrative use: highlight call patterns and interactions that need review.

Customer: “I’ve contacted support three times already!”

ASR

Transcribe the spoken words.

Emotion model

Estimate possible frustration cues.

AI analysis

Interpret combined outputs with context.

Future agentic extension: check case history → decide next steps → escalate with approval.

RTMT: Meeting Support and Next Steps

Real-Time Meeting Translation

Current capabilities

Live translation, transcripts, and post-meeting minutes.

Potential extension

Use meeting outputs to help track project tasks and progress.

Illustrative workflow

Find action items → propose task updates → request approval.

Project tracking is a potential extension; translation and meeting outputs are available today.

Speech-to-Speech AI: Company Knowledge by Voice

Voice question → Company knowledge → Spoken answer

Current capabilities

Answer voice queries using internal data and documents.

Connection to agents

A voice interface with retrieved knowledge can support an agent.

Potential extension

Add permitted tools to carry out and verify requested tasks.

Illustrative extension: answer a policy question, then help submit a request with approval.

Product Demo: Cartesia

An AI voice assistant that speaks three languages, answers customer questions, and automatically saves call records.

https://drive.google.com/file/d/1Vof27JqdRYwh2GH731z0NkjKmjXoCUfI/view?usp=sharing

What does a Data Scientist do in Speech AI?

Problem
Define the real need

Data
Prepare and understand recordings

Experiment
Build and compare models

Evaluate
Measure performance and errors

Improve
Refine and validate

A large part of the work happens outside model training:

Understand the data
Labels, speakers, recording conditions

Investigate errors
Where and why does it fail?

Communicate
Explain trade-offs and risks

Agentic AI in Our Development Workflow

1 Data Checks

Support quality checks before model training

2 Experiments

Assist with planned model comparisons

3 Result Analysis

Help summarize metrics and investigate errors

4 Human Review

Our team validates findings and approves next steps

Used to support experiments and data quality—not to replace validation by our team.

High accuracy is not the whole story

High accuracy is valuable, but context and error impact still matter.

Illustrative examples

BelugaBox / EmoSDK

- An excited voice is confidently labeled “angry”.
- Emotion is subjective; call context may suggest a different interpretation.
- Use emotion scores as clues, not facts.

RTMT

- “We might finish Friday” becomes “We will finish Friday.”
- Most words are correct, but the commitment changes.
- Confirm intent before updating a project plan.

Speech-to-Speech AI

- The words are recognized correctly, but the answer uses an outdated policy.
- Correct words can still lead to a wrong answer or action.
- Check current sources before acting.

Evaluate context, uncertainty, and the consequences of errors.

When recognition errors become actions

A small error can have a large impact when an agent acts.

Without verification

- Hears “fifteen” as “fifty”
- Books 50 seats without checking
- An ASR error becomes a wrong booking

With verification

- Checks a critical booking detail
- Asks: “Did you mean 15 seats?”
- Confirms before booking

Evaluate the whole system:

Real-world audio	Critical errors	Handling uncertainty	Task success

An agent’s decisions depend on recognition quality—and checks before taking action.

Challenges in real-world voice agents

Audio quality

Noise, microphones, distance

Language variation

Accents, dialects, pronunciation

Meaning & context

Ambiguity, sarcasm, intent

Wrong actions

Misheard details or bad decisions

Latency

Users expect fast responses

Privacy & safety

Sensitive voice data and permissions

A good system is not only intelligent — it must also be reliable, safe, and useful.

Where could voice AI go next?

More natural	Real-time, fluid conversations
More inclusive	Multilingual and accent-aware systems
More contextual	Better use of conversation history
More capable	Agents that coordinate tools and tasks
More responsible	Privacy, transparency, human oversight

Thank you • Q&A

If you could build a voice agent, what would you want it to do?

Mentimeter Slide 5

University assistant
Answer campus questions

Study planner
Organize assignments and time

Programming tutor
Explain and debug code

Language partner
Practice speaking

Key takeaways

Speech Recognition helps AI hear us. Agentic AI helps AI do something with what it hears.

NEXT: HANDS-ON WORKSHOP

Now, let's build a voice agent

Speak a request. See the action. Hear the reply.

1. Get together in groups of 3.
2. Explore the notebook together and share your ideas.
3. Submit your group's work using the Google Form.

Open the hands-on guide